

GAE: General Action Expert for Real-Time Humanoid Teleoperation

Yuefan Wang^{1,2} Huaicheng Zhou¹ Xiao He¹ Zhijie He¹ Mingchuan Yang¹
Huayi Zhang¹ Li Chai² Jinxin Liu^{1,*} Donglin Wang^{1,2,*}

¹Westlake Robotics ²Westlake University *Corresponding authors

Project Website: <https://wangyf0928.github.io/gae-wlrobotics/>

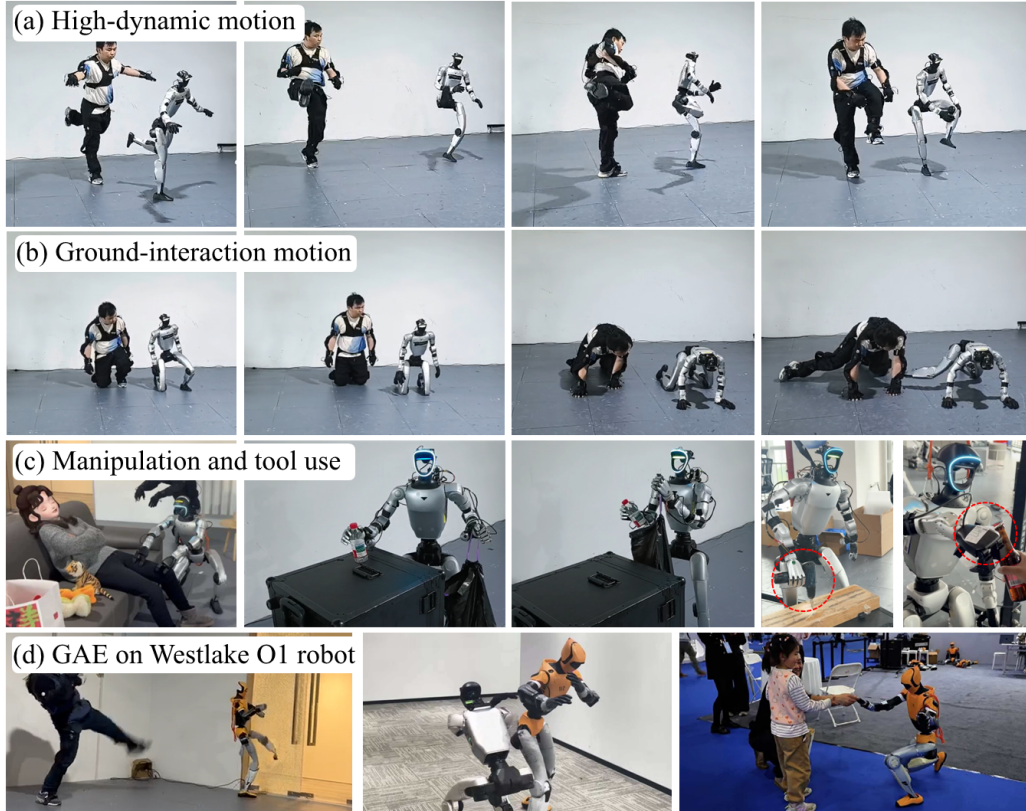


Figure 1: **Real-time whole-body humanoid teleoperation with GAE.** GAE enables humanoid robots to act as responsive physical avatars of human users across diverse motions, tasks, and hardware. (a) High-dynamic motions. (b) Ground-interaction motions. (c) Manipulation and tool use with Unitree Dex3 three-finger hands and BrainCo Revo2 dexterous hands. (d) Cross-platform deployment on the Westlake O1 robot.

Abstract: Humanoid avatars extend human physical presence beyond the body, enabling people to participate in social, service, and labor activities through remotely operated robots. This requires teleoperation systems capable of realizing diverse and dynamic whole-body behaviors while maintaining responsive human-robot synchronization. We present **General Action Expert (GAE)**, a unified learning framework for general-purpose, low-latency humanoid whole-body teleoperation. To cover diverse human behaviors, GAE builds a large-scale human motion dataset from heterogeneous sources, including videos, animations, and motion capture, followed by standardization and augmentation. GAE then addresses the noise and embodiment mismatch in human motions with a two-stage

training paradigm: a privileged generator policy first tracks human motion references in simulation and rolls out feasible humanoid trajectories; a deployable executor policy then learns to track these generated trajectories under curriculum domain randomization. For responsive human-robot synchronization, GAE introduces a latency-conditioned anticipation mechanism that adaptively compensates for end-to-end delay during real-time teleoperation. Simulation and real-world experiments on Unitree G1 and Westlake O1 robots demonstrate that GAE enables humanoids to smoothly mirror diverse, agile, and expressive human behaviors.

Keywords: Humanoid Robot, Whole-body Teleoperation, Embodied Intelligence

1 Introduction

Owing to their human-like morphology, humanoid robots are naturally compatible with human-centric environments and can reproduce a wide range of human behaviors [1, 2, 3, 4]. These properties make humanoids a compelling platform for building physical avatars, where a teleoperated robot extends a user’s presence and agency beyond their own body [5, 6, 7, 8]. Such avatars could enable people to participate in social interaction, service tasks, and physical labor in remote, hazardous, or otherwise inaccessible environments without being physically present.

Realizing this vision imposes two fundamental requirements on the teleoperation system. First, the robot must support diverse, agile, and expressive whole-body behaviors, as human operators naturally perform open-ended motions during teleoperation. Second, the robot must remain tightly synchronized with the operator, as even small delays between human motion and robot response can weaken controllability and reduce the sense of embodiment in real-time teleoperation [9, 10].

The first requirement, broad whole-body motion coverage, has motivated recent efforts in reinforcement-learning-based humanoid control [11]. A key prerequisite for such diversity is access to large-scale human motion data [12, 13]. However, high-quality motion-capture data remain limited in scale, while broader sources such as videos and character animations often contain substantial noise and ambiguity. In addition, human motions cannot be used directly as humanoid control targets due to the mismatch between human bodies and robot morphologies [14]. Kinematic retargeting can map human motions to humanoid-compatible references, but the resulting trajectories may be unstable, discontinuous, or physically inconsistent, introducing noisy supervision for policy learning. Besides, retargeting introduces an additional processing step during real-time deployment, which increases system complexity and latency [15]. More critically, deployable policies often rely on extensive domain randomization to bridge the sim-to-real gap [16], which introduces complex and stochastic physical constraints into training. The compounding effect of noisy motion targets and randomized physical constraints makes policy learning unstable and difficult to scale to diverse and dynamic whole-body behaviors.

The second requirement, responsive human-robot synchronization, presents a different but equally critical challenge, yet is less commonly treated as an explicit design objective in humanoid whole-body control works [17, 18, 19]. In practical deployment, delays are inevitably introduced by sensing, communication, preprocessing, policy inference, and low-level actuation. These delays cause the robot to react to outdated human motion signals, creating a visible temporal gap between the operator and the humanoid avatar, which is particularly detrimental for motions requiring precise timing and responsiveness.

In this work, we present **General Action Expert (GAE)**, a unified learning framework for general-purpose, low-latency humanoid whole-body teleoperation. GAE targets the two challenges above: scaling humanoid control to open-ended human behaviors and maintaining responsive synchronization under real-world latency. To this end, GAE integrates three key components: a large-scale human motion dataset, a two-stage training paradigm that decouples noisy motion targets from randomized physical constraints, and a latency-conditioned anticipation mechanism.

To support broad motion diversity, GAE first builds a large-scale human motion dataset from heterogeneous sources, including videos, character animations, and motion-capture data. All motions are standardized and further expanded through three augmentation strategies: mirroring, which generates left-right variants of existing motions; stitching, which connects motion sequences with similar transitional poses; and recombination, which recombines upper- and lower-body motions from different sequences. These procedures substantially expand the coverage of human behaviors while preserving a coherent whole-body structure.

On top of this data set, GAE adopts a two-stage training paradigm to enable effective large-scale policy learning for humanoid teleoperation. The key idea is to first ensure reliable tracking references and then learn sim-to-real robustness for deployment. In the first stage, human motion references are geometrically adapted to the humanoid scale through shape fitting. A privileged generator policy is then trained in a clean simulation setting to track these shape-fitted references. By rolling out this policy, GAE converts noisy and embodiment-mismatched human motions into physically feasible humanoid trajectories with accuracy and stability. In the second stage, a deployable executor policy learns to track these generated trajectories under curriculum domain randomization. Importantly, we explicitly condition the executor on raw human motion observations rather than the generated humanoid trajectories. As a result, the deployed policy can directly react to human motions without relying on any intermediate retargeting or trajectory-generation module. This design naturally separates motion denoising and embodiment adaptation from sim-to-real robustness learning. By avoiding the coupling between noisy human-motion references and randomized physical dynamics, the framework enables more stable and effective large-scale policy optimization.

To further improve teleoperation responsiveness, GAE incorporates a latency-conditioned anticipation mechanism. Instead of passively reacting to the latest available human motion, the executor learns to predictively track future human motion states based on the observed motion context. During deployment, the horizon of future anticipation can be adjusted according to the end-to-end latency of the teleoperation pipeline, allowing the policy to proactively compensate for communication and processing delays. As a result, the humanoid avatar maintains tighter temporal alignment with the operator during real-time teleoperation.

Together, these designs form a unified framework that advances humanoid teleoperation by enabling effective policy learning and latency-aware motion control. We evaluate GAE in both simulation and real-world deployment on Unitree G1 and Westlake O1 humanoid robots. As shown in Fig. 1, GAE enables humanoids to smoothly mirror diverse whole-body behaviors, including dynamic locomotion, contact-rich ground behaviors, tool-use manipulation, and loco-manipulation. These results demonstrate the potential of GAE for building responsive humanoid avatars capable of robust real-time teleoperation across diverse platforms and motion scenarios.

2 Related Work

Humanoid Teleoperation. Humanoid teleoperation offers an intuitive way to transfer human intent to robots and is essential for real-world humanoid deployment. Existing systems use various input modalities, including sparse VR keypoints, exoskeletons [20], motion-capture suits, and camera-based pose estimation [21]. Many of them, however, decouple upper- and lower-body control, with human commands controlling the arms or torso and a separate locomotion policy handling walking, which limits coordinated whole-body behaviors such as kicking, crouching, stepping, and legged manipulation. Recent works train policies to track human motions for whole-body teleoperation, but camera-based systems remain sensitive to occlusion and root-motion errors, while sparse VR inputs only partially observe the body. Although motion capture provides more accurate whole-body signals, it still requires a controller that can produce dynamically feasible robot actions with minimal latency. GAE addresses this by learning a unified whole-body controller directly conditioned on human motion observations, avoiding an additional online retargeting stage.

Learning-based Humanoid Whole-body Control. Learning-based humanoid control has evolved from task-specific locomotion policies to general whole-body motion tracking controllers. Recent

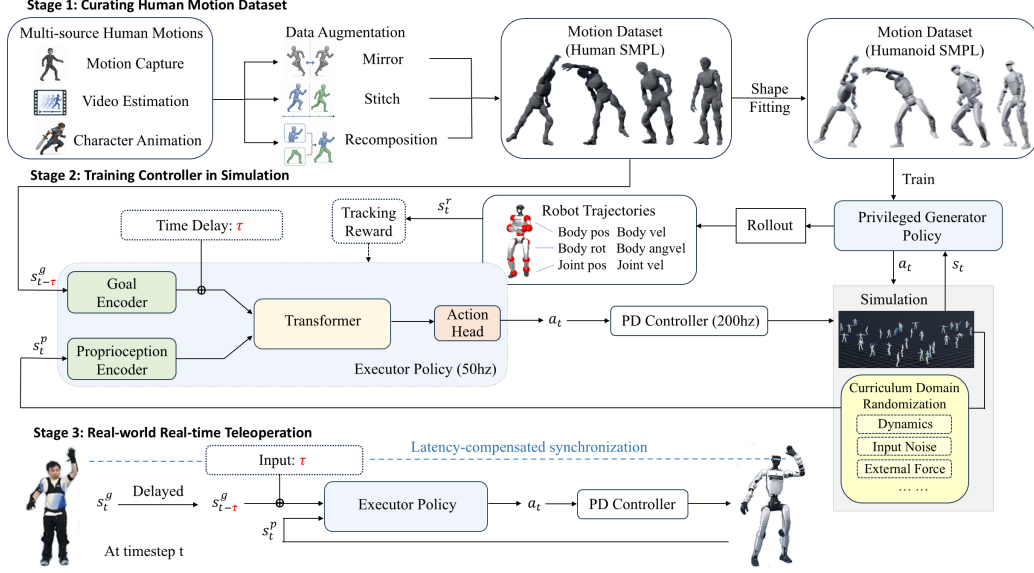


Figure 2: **Overview of GAE framework.** GAE consists of three stages: 1) curating a large-scale multi-source human motion dataset, 2) training the controller in simulation through a generator-executor paradigm, 3) teleoperating real-world robots with latency-conditioned synchronization.

works improve the generality of humanoid control by learning from large-scale human motion data [22, 23, 24, 25, 26]. Despite some progress, robustly tracking open-ended human behaviors remains challenging. Moreover, temporal responsiveness under real-world teleoperation latency is rarely treated as an explicit controller-level objective. In contrast, our work develops a general-purpose whole-body controller that supports broader dynamic behaviors while explicitly addressing low-latency real-world teleoperation.

3 Methods

Problem Formulation. We formulate humanoid whole-body teleoperation as a goal-conditioned reinforcement learning problem. At each control step t , the policy π receives the robot proprioceptive state s_t^p and a human-motion goal s_t^g , and outputs an action $a_t \in \mathbb{R}^{\text{DOF}}$, where DOF denotes the number of robot joints. The action a_t specifies target joint positions that are executed by a low-level PD controller. The policy is optimized with Proximal Policy Optimization (PPO) [27] to maximize the expected discounted return $\mathbb{E} \left[\sum_{t=1}^T \gamma^{t-1} r_t \right]$, where the reward $r_t = R(s_t^p, s_t^g)$ provides dense whole-body motion-tracking feedback.

Unlike standard motion-tracking settings where the reference motion is assumed to be instantaneously available, real-world teleoperation inevitably introduces end-to-end latency from sensing, communication, preprocessing, policy inference, and actuation. Consequently, at time t , the robot receives a delayed human-motion command $s_{t-\tau}^g$, which reflects the operator’s intent τ timesteps in the past rather than the current intent s_t^g . This temporal discrepancy makes it difficult to maintain synchronization between the operator and the humanoid through direct goal tracking alone. To address this challenge, GAE explicitly conditions the policy on the system latency τ together with the latest available human-motion command $s_{t-\tau}^g$. Crucially, although the policy observes only the delayed command, its tracking reward is evaluated with respect to the current human intent s_t^g . In this way, the policy is trained to map delayed intent observations to real-time tracking behavior: taking $(s_{t-\tau}^g, \tau)$ as input, it must produce actions that best match the operator’s intent s_t^g . This formulation encourages the network to internally compensate for latency, leading to tighter human-robot synchronization and more responsive teleoperation during deployment.

Humanoid Motion Dataset. To support diverse and expressive whole-body teleoperation, we build a large-scale human motion dataset from multiple sources, including motion-capture datasets [28], video-estimated motions, and character animation data. We convert all motions into a unified SMPL-based human body representation and enrich the dataset through symmetry-based mirroring, temporal stitching, and upper-lower body recombination across different motion sequences. We further construct a humanoid-proportioned SMPL [29] body model as a geometric proxy for the robot. The model matches the body scale and limb proportions of the humanoid robot while retaining the spherical-joint kinematic structure of SMPL, rather than directly adopting the robot’s hinge-joint morphology. We then fit the collected human motions to this proxy skeleton, producing morphology-aligned motion references that provide training targets for the generator policy.

Generator Policy Training. The generator policy is responsible for producing physically feasible robot motion trajectories with accuracy and stability that serve as tracking target for the executor policy. During training, we learn the generator policy $\pi^{\text{gen}}(\cdot \mid s_t^{\text{p-gen}}, s_t^{\text{g-gen}})$ in a clean simulation setting, *i.e.*, it has access to privileged information and is not exposed to any environment disturbances. Formally, the proprioception state is defined as $s_t^{\text{p-gen}} \triangleq [\mathbf{p}_t^{\text{loc}}, \dot{\mathbf{p}}_t, \boldsymbol{\omega}_t, \mathbf{f}_t, \mathbf{r}_t, \boldsymbol{\omega}_t^{\text{root}}, \mathbf{q}_t, \dot{\mathbf{q}}_t, \mathbf{a}_{t-1}]$, which contains the local body position $\mathbf{p}_t^{\text{loc}}$, linear velocity $\dot{\mathbf{p}}_t$, angular velocity $\boldsymbol{\omega}_t$, contact force $\mathbf{f}_t$, root orientation $\mathbf{r}_t$, root angular velocity $\boldsymbol{\omega}_t^{\text{root}}$, joint position $\mathbf{q}_t$, joint velocity $\dot{\mathbf{q}}_t$, and the previous actions $\mathbf{a}_{t-1}$. The goal state is defined as $s_t^{\text{g-gen}} \triangleq [\hat{\mathbf{p}}_t^{\text{loc}}, \hat{\mathbf{v}}_t, \hat{\mathbf{r}}_t]$, which contains local body positions, linear velocities, and body rotations of reference motion. The complete observation provided to the generator policy consists of a 100-frame history of both proprioceptive and goal states, together with an 32-frame future horizon of goal states. After training the generator policy, we collect robot motion trajectories by rolling it out in simulation. Each trajectory records the robot’s body states $\{s_t^r\}$, including body positions, linear/angular velocities, rotations, joint positions, and joint velocities.

Executor Policy Training. Given the clean and feasible motion targets $\{s_t^r\}$ collected from the generator policy, the executor policy aims to learn a general and responsive controller that can be directly deployed for real-world humanoid teleoperation. To bridge the sim-to-real gap, we train the executor policy with curriculum domain randomization, where the strength and diversity of domain randomization are progressively increased during training. This curriculum gradually exposes the policy to variations in robot dynamics, observation noise, and external perturbations, allowing it to first acquire stable motion-tracking behaviors and then improve robustness under increasingly challenging conditions. Importantly, the generator trajectories are not exposed to the executor policy as observations; they only serve as tracking rewards to shape the learning objective. The executor policy instead observes raw human motion inputs. Consequently, at deployment time, the executor policy can react directly to human motion inputs without relying on retargeted robot trajectories as intermediate commands. Formally, the proprioception state of the executor policy is defined as $s_t^{\text{p}} \triangleq [\mathbf{g}_t, \boldsymbol{\omega}_t^{\text{root}}, \mathbf{q}_t, \dot{\mathbf{q}}_t, \mathbf{a}_{t-1}]$, which includes 100-step history of robot state. The goal state is $s_t^{\text{g}} \triangleq [\hat{\mathbf{p}}_t^{\text{h,loc}}, \hat{\mathbf{v}}_t^{\text{h}}, \hat{\mathbf{r}}_t^{\text{h}}]$, which represents the original human motion intents. The reward is computed based on the rollout state s_t^r recorded in the generated trajectory.

Latency-conditioned Anticipation. To further improve real-time synchronization, we introduce a latency-conditioned anticipation mechanism for the executor policy. In practical teleoperation, the human motion received by the robot is inevitably delayed. To compensate for this delay, we explicitly condition the executor policy $\pi^{\text{exe}}(\cdot \mid s_t^{\text{p}}, s_{t-\tau}^{\text{g}}, \tau)$ on a time delay τ , allowing it to track human motion states predictively rather than mimicking the delayed motion observation itself. Specifically, we tokenize the raw human-motion observations together with the robot proprioceptive observations and encode their temporal positions using rotary position embeddings (RoPE) [30]. The time delay, which we also refer to as the policy anticipation horizon, is represented by shifting the RoPE indices of the human-motion tokens. This shift changes the temporal motion frame interpreted by the policy: a larger anticipation horizon indicates that the policy should track a farther future motion state based on the current human-motion observations. During training, the anticipation horizon is randomly sampled across parallel environments, so that a single executor policy can adapt to different deployment delays. During deployment, this horizon can be adjusted according to the measured latency of the teleoperation pipeline.

Network Architecture. Both the generator policy and the executor policy use two separate MLP encoders to encode the proprioceptive state and the goal state, respectively. The encoded features are then processed by a mixture of expert Transformer [31] backbone, and the temporal position embeddings are applied in every block of the backbone. Finally, an MLP action head maps the last output token to the action a_t . We train our policies in Isaac Lab [32] with parallel environments.

Reward Design and Domain Randomization. The generator policy and executor policy share a similar reward design, differing primarily in their reference motions. For the generator policy, the tracking reward is computed with respect to the shape-fitted human motion, while for the executor policy, the tracking reward is computed with respect to the trajectories rolled out by the generator policy. We formulate the reward r_t as the sum of two components: 1) penalty, and 2) task rewards for motion tracking. We apply extensive curriculum domain randomization on the executor policy training for successful sim-to-real transfer. The details of the parameters for reward designs and domain randomization are provided in the Appendix A.

4 Experiments

In this section, we conduct extensive experiments to answer the following questions: **(1)** Can GAE accurately track diverse human motions? **(2)** Can GAE maintain strong tracking performance under different anticipation horizons for latency compensation? **(3)** Can GAE support robust open-world teleoperation? **(4)** Can the GAE training framework transfer to different humanoid embodiments?

Experimental Setup and Metrics. To evaluate the model under different levels of motion complexity, we construct three evaluation motion sets, denoted as Easy, Medium, and Hard. These sets contain diverse whole-body human motions with progressively increasing dynamics. The evaluation on these three sets assesses the robustness and generalization across motion difficulty levels. We report four tracking errors over selected body keypoints: root-relative Mean Keypoint Position Error(MKPE), Mean Keypoint Rotation Error(MKRE), Mean Keypoint Linear Velocity Error(MKLVE), and Mean Keypoint Angular Velocity Error(MKAVE). These metrics measure whether the robot can accurately reproduce the reference motion in both pose and velocity space. In addition, we report a motion tracking Success Rate(SR). A motion sequence is considered successfully tracked only if every tracked keypoint remains within 50 cm of the reference trajectory. Compared with average tracking errors, SR better captures complete-sequence robustness by penalizing transient large deviations that may cause loss of synchronization.

Table 1: Quantitative results.

Method	SR (%) $\uparrow$	MKPE $\downarrow$	MKLVE $\downarrow$	MKAVE $\downarrow$	MKRE $\downarrow$
(a) Comparison with SONIC on Different Test Sets					
<i>Easy Set</i>					
SONIC	95.4	0.036	0.271	0.923	0.224
GAE	97.3	0.033	0.244	1.382	0.275
<i>Medium Set</i>					
SONIC	85.2	0.045	0.437	1.385	0.278
GAE	96.9	0.046	0.263	1.076	0.269
<i>Hard Set</i>					
SONIC	77.7	0.043	0.595	1.489	0.367
GAE	93.9	0.045	0.284	1.443	0.347
(b) Performance under Different Anticipation Horizons					
$\tau = 0$	93.9	0.045	0.241	1.443	0.347
$\tau = 1$	93.7	0.046	0.252	1.512	0.355
$\tau = 2$	94.0	0.048	0.270	1.598	0.365
$\tau = 3$	93.8	0.050	0.290	1.692	0.376
$\tau = 4$	92.8	0.053	0.309	1.764	0.389
$\tau = 5$	92.1	0.056	0.326	1.837	0.403

4.1 Simulation Motion Tracking Results

We first compare GAE with SONIC, a representative and currently state-of-the-art open-source whole-body controller for humanoid robots. As shown in Table 1(a), GAE achieves performance comparable to SONIC on the Easy set, where both methods can reliably track relatively simple whole-body motions. The advantage of GAE becomes more pronounced as the motion difficulty increases. GAE consistently achieves better overall performance on the Medium and Hard sets. In particular, GAE maintains higher sequence-level success rates as motion dynamics increase, showing stronger robustness for challenging whole-body motions. These results indicate that GAE can reliably support a broader range of human motions, especially highly dynamic ones.

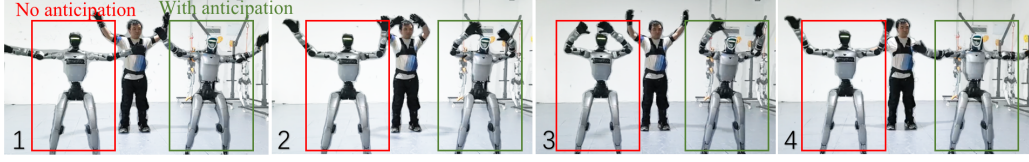


Figure 3: Synchronization evaluation in real world



Figure 4: Open-ended fight performance through teleoperation

We further evaluate whether GAE can support deployment-time latency compensation through adaptive anticipation horizons. Since the controller runs at 50 Hz, advancing the tracking target by one frame corresponds to 20 ms of anticipation. Thus, anticipation horizons from $\tau = 0$ to $\tau = 5$ provide 0–100 ms of lead time, enabling the same policy to compensate for system latency up to 100 ms. However, anticipating further into the future makes motion tracking more challenging and may reduce tracking accuracy. To study this trade-off, we evaluate the same trained GAE model under different anticipation horizons on the hard set, where highly dynamic motions impose the greatest challenge for anticipation-based control. Importantly, no model is retrained for different horizons; we only adjust the horizon at evaluation time through the temporal position embedding.

As shown in Table 1(b), increasing the anticipation horizon leads to a gradual but moderate degradation in tracking precision. As τ increases from 0 to 5, MKPE increases from 0.045 to 0.056, and the velocity- and rotation-related errors also grow consistently. This trend is expected, since a larger horizon requires the policy to track farther states and makes anticipation-based control more challenging, especially on the hard set with highly dynamic motions. Nevertheless, the overall performance remains stable across all horizons. Even with $\tau = 5$, which corresponds to 100 ms of anticipation, GAE still achieves a success rate of 92.1% and maintains relatively low tracking errors. These results indicate that GAE can trade a small amount of tracking precision for deployment-time latency compensation, while still robustly executing challenging whole-body motions.

4.2 Real-world Motion Tracking Results

Latency compensation through anticipation. In real-world deployment, we first evaluate whether the anticipation mechanism of GAE can reduce the visible delay between the human operator and the robot. As shown in Fig. 3, we let two Unitree G1 robots track the same motion-capture operator simultaneously. Both robots are controlled by GAE, but they use different anticipation horizons. The first robot uses $\tau = 0$, which means that it directly tracks the currently available human motion without compensation. The second robot uses $\tau = 4$, corresponding to an 80 ms anticipation horizon under our 50 Hz control frequency. The comparison shows a clear difference in motion synchronization. Without anticipation, the robot exhibits an observable lag behind the operator: when the operator has already moved to a new pose, the robot is still reproducing an earlier pose. In contrast, the robot using $\tau = 4$ remains more synchronized with the human motion throughout the sequence. This result demonstrates the effectiveness of the proposed anticipation mechanism, showing that explicitly conditioning the policy on the anticipation horizon improves temporal synchronization between the operator and the humanoid robot.

Open-world teleoperation. Beyond laboratory demonstrations, we deploy GAE in a public open-world setting to showcase its practical capability for real-time humanoid teleoperation. Specifically,



Figure 5: Cross-platform deployment on Westlake O1 robot

GAE was deployed in a publicly held live performance event. As shown in Fig. 4, two Unitree G1 robots controlled by GAE are teleoperated by two motion-capture users in a live fighting performance. Unlike controlled laboratory experiments, this scenario involves unscripted operator behaviors, rapidly changing whole-body motions, and long-duration execution in a realistic environment.

This deployment demonstrates that GAE can support diverse and robust humanoid behaviors outside controlled benchmark settings. The robots reproduce a wide range of operator motions, including fast arm swings, body turns, stepping, and kicking, while maintaining stable control throughout the demonstration. These results suggest that GAE can move beyond isolated scripted motion tracking and serve as a practical controller for open-world real-time humanoid teleoperation.

Transfer to other humanoid embodiment. Finally, we investigate whether the GAE training framework can be transferred to a different humanoid platform. In addition to Unitree G1, we deploy GAE on the Westlake O1 robot, which has a similar overall humanoid morphology but different embodiment details. We fine-tune the GAE controller on the O1 platform and evaluate it with diverse human motions. As shown in Fig. 5, the finetuned controller enables the Westlake O1 robot to perform a broad range of behaviors. This experiment shows that our method is not tied to a single robot embodiment. Instead, it provides a transferable framework for adapting human-motion-driven whole-body control to different humanoid robots.

5 Limitations and Future Work

Although GAE enables general and low-latency whole-body teleoperation, it still has several limitations. First, the current controller does not incorporate external perception, such as RGB-camera or LiDAR. As a result, the policy mainly responds to human motion commands and proprioceptive states, but lacks direct awareness of the surrounding environment. This limits its adaptability in complex terrains or contact-rich locomotion scenarios, such as ascending stairs and climbing upwards. Second, GAE does not yet integrate tactile sensors. The controller receives no direct tactile sensing from the robot, and the operator is not provided with force feedback during teleoperation. Without tactile feedback, the operator may have limited awareness of contact states, while the policy may not sufficiently adapt its actions to subtle force variations. This makes the system less suitable for manipulation tasks that require accurate force regulation, such as handling fragile objects, or interacting safely with delicate environments. Thus, we plan to extend GAE with richer external perception inputs in future work to improve the robot’s understanding of its environment and enable more adaptive whole-body control in challenging terrains and cluttered scenes. Meanwhile, we also plan to integrate tactile sensing and force feedback to enhance fine-grained manipulation, contact safety, and operator awareness. We believe these extensions will lead to a more general, capable, and robust whole-body humanoid teleoperation system.

References

- [1] X. Cheng, Y. Ji, J. Chen, R. Yang, G. Yang, and X. Wang. Expressive whole-body control for humanoid robots. *arXiv preprint arXiv:2402.16796*, 2024.
- [2] Z. Chen, M. Ji, X. Cheng, X. Peng, X. B. Peng, and X. Wang. Gmt: General motion tracking for humanoid whole-body control. *arXiv preprint arXiv:2506.14770*, 2025.
- [3] Y. Wang, S. Zhu, P. Zhi, Y. Li, J. Li, Y.-L. Li, Y. Xiao, X. Wang, B. Jia, and S. Huang. Omnixtreme: Breaking the generality barrier in high-dynamic humanoid control. *arXiv preprint arXiv:2602.23843*, 2026.
- [4] W. Xie, J. Han, J. Zheng, H. Li, X. Liu, J. Shi, W. Zhang, C. Bai, and X. Li. Kungfubot: Physics-based humanoid whole-body control for learning highly-dynamic skills. *Advances in Neural Information Processing Systems*, 38:62406–62433, 2026.
- [5] Y. Li, L. Ma, Y. Lin, Y. Du, M. Liu, K. Hu, J. Cui, Y. Zhu, W. Liang, B. Jia, et al. Omniclone: Engineering a robust, all-rounder whole-body humanoid teleoperation system. *arXiv preprint arXiv:2603.14327*, 2026.
- [6] N. Myers, O. Kwon, S. Yamsani, and J. Kim. Child (controller for humanoid imitation and live demonstration): A whole-body humanoid teleoperation system. In *2025 IEEE-RAS 24th International Conference on Humanoid Robots (Humanoids)*, pages 1–6. IEEE, 2025.
- [7] T. Zhu, G. Cai, Y. Zhaohui, G. Ren, H. Xie, Z. Wang, J. Wu, J. Wang, X. Yang, Y. Mu, et al. Clot: Closed-loop global motion tracking for whole-body humanoid teleoperation. *arXiv preprint arXiv:2602.15060*, 2026.
- [8] K. Darvish, L. Penco, J. Ramos, R. Cisneros, J. Pratt, E. Yoshida, S. Ivaldi, and D. Pucci. Teleoperation of humanoid robots: A survey. *IEEE Transactions on Robotics*, 39(3):1706–1727, 2023.
- [9] Z. Xiong, L. Fang, J. Huang, K. Yamazaki, H. Zhang, and C. Gan. Extremcontrol: Low-latency humanoid teleoperation with direct extremity control. *arXiv preprint arXiv:2602.11321*, 2026.
- [10] S. B. Kamtam, Q. Lu, F. Bouali, O. C. Haas, and S. Birrell. Network latency in teleoperation of connected and autonomous vehicles: A review of trends, challenges, and mitigation strategies. *Sensors*, 24(12):3957, 2024.
- [11] Y. Wang, M. Yang, G. Ding, Y. Zhang, W. Zeng, X. Xu, H. Jiang, and Z. Lu. From experts to a generalist: Toward general whole-body control for humanoid robots. *Advances in Neural Information Processing Systems*, 38:147748–147772, 2026.
- [12] J. Lin, A. Zeng, S. Lu, Y. Cai, R. Zhang, H. Wang, and L. Zhang. Motion-x: A large-scale 3d expressive whole-body human motion dataset. *Advances in Neural Information Processing Systems*, 36:25268–25280, 2023.
- [13] F. G. Harvey, M. Yurick, D. Nowrouzezahrai, and C. Pal. Robust motion in-betweening. 39 (4), 2020.
- [14] J. P. Araujo, Y. Ze, P. Xu, J. Wu, and C. K. Liu. Retargeting matters: General motion retargeting for humanoid motion tracking. *arXiv preprint arXiv:2510.02252*, 2025.
- [15] Y. Ze, Z. Chen, J. P. Araújo, Z. ang Cao, X. B. Peng, J. Wu, and C. K. Liu. Twist: Teleoperated whole-body imitation system. *arXiv preprint arXiv:2505.02833*, 2025.
- [16] Z. Zhang, J. Guo, C. Chen, J. Wang, C. Lin, Y. Lian, H. Xue, Z. Wang, M. Liu, J. Lyu, et al. Track any motions under any disturbances. *arXiv preprint arXiv:2509.13833*, 2025.
- [17] Z. Fu, Q. Zhao, Q. Wu, G. Wetzstein, and C. Finn. Humanplus: Humanoid shadowing and imitation from humans. *arXiv preprint arXiv:2406.10454*, 2024.

- [18] T. He, Z. Luo, W. Xiao, C. Zhang, K. Kitani, C. Liu, and G. Shi. Learning human-to-humanoid real-time whole-body teleoperation. In *2024 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, pages 8944–8951. IEEE, 2024.
- [19] T. He, Z. Luo, X. He, W. Xiao, C. Zhang, W. Zhang, K. Kitani, C. Liu, and G. Shi. Omnih2o: Universal and dexterous human-to-humanoid whole-body teleoperation and learning. *arXiv preprint arXiv:2406.08858*, 2024.
- [20] R. Zhong, C. Cheng, J. Xu, Y. Wei, C. Guo, D. Zhang, W. Dai, and H. Lu. Nuexo: A wearable exoskeleton covering all upper limb rom for outdoor data collection and teleoperation of humanoid robots. In *2025 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, pages 12026–12033. IEEE, 2025.
- [21] J. Li, L. Yang, and A. D. Ames. Mirror: Visual motion imitation via real-time retargeting and teleoperation with parallel differential inverse kinematics. *arXiv preprint arXiv:2603.23995*, 2026.
- [22] M. Ji, X. Peng, F. Liu, J. Li, G. Yang, X. Cheng, and X. Wang. Exbody2: Advanced expressive humanoid whole-body control. *arXiv preprint arXiv:2412.13196*, 2024.
- [23] T. He, W. Xiao, T. Lin, Z. Luo, Z. Xu, Z. Jiang, J. Kautz, C. Liu, G. Shi, X. Wang, et al. Hover: Versatile neural whole-body controller for humanoid robots. In *2025 IEEE International Conference on Robotics and Automation (ICRA)*, pages 9989–9996. IEEE, 2025.
- [24] K. Yin, W. Zeng, K. Fan, M. Dai, Z. Wang, Q. Zhang, Z. Tian, J. Wang, J. Pang, and W. Zhang. Unitracker: Learning universal whole-body motion tracker for humanoid robots. *IEEE Robotics and Automation Letters*, 2026.
- [25] Q. Liao, T. E. Truong, X. Huang, Y. Gao, G. Tevet, K. Sreenath, and C. K. Liu. Beyondmimic: From motion tracking to versatile humanoid control via guided diffusion. *arXiv preprint arXiv:2508.08241*, 2025.
- [26] Z. Luo, Y. Yuan, T. Wang, C. Li, S. Chen, F. Castaneda, Z.-A. Cao, J. Li, D. Minor, Q. Ben, et al. Sonic: Supersizing motion tracking for natural humanoid whole-body control. *arXiv preprint arXiv:2511.07820*, 2025.
- [27] J. Schulman, F. Wolski, P. Dhariwal, A. Radford, and O. Klimov. Proximal policy optimization algorithms. *arXiv preprint arXiv:1707.06347*, 2017.
- [28] N. Mahmood, N. Ghorbani, N. F. Troje, G. Pons-Moll, and M. J. Black. AMASS: Archive of motion capture as surface shapes. In *International Conference on Computer Vision*, pages 5442–5451, Oct. 2019.
- [29] M. Loper, N. Mahmood, J. Romero, G. Pons-Moll, and M. J. Black. Smpl: A skinned multi-person linear model. In *Seminal Graphics Papers: Pushing the Boundaries, Volume 2*, pages 851–866. 2023.
- [30] J. Su, M. Ahmed, Y. Lu, S. Pan, W. Bo, and Y. Liu. Roformer: Enhanced transformer with rotary position embedding. *Neurocomputing*, 568:127063, 2024.
- [31] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, L. u. Kaiser, and I. Polosukhin. Attention is all you need. In I. Guyon, U. V. Luxburg, S. Bengio, H. Wallach, R. Fergus, S. Vishwanathan, and R. Garnett, editors, *Advances in Neural Information Processing Systems*, volume 30. Curran Associates, Inc., 2017. URL https://proceedings.neurips.cc/paper_files/paper/2017/file/3f5ee243547dee91fbd053c1c4a845aa-Paper.pdf.

- [32] M. Mittal, P. Roth, J. Tigue, A. Richard, O. Zhang, P. Du, A. Serrano-Muñoz, X. Yao, R. Zurbrugg, N. Rudin, L. Wawrzyniak, M. Rakhsha, A. Denzler, E. Heiden, A. Borovicka, O. Ahmed, I. Akinola, A. Anwar, M. T. Carlson, J. Y. Feng, A. Garg, R. Gasoto, L. Gulich, Y. Guo, M. Gussert, A. Hansen, M. Kulkarni, C. Li, W. Liu, V. Makoviychuk, G. Malczyk, H. Mazhar, M. Moghani, A. Murali, M. Noseworthy, A. Poddubny, N. Ratliff, W. Rehberg, C. Schwarke, R. Singh, J. L. Smith, B. Tang, R. Thaker, M. Trepte, K. V. Wyk, F. Yu, A. Millane, V. Ramasamy, R. Steiner, S. Subramanian, C. Volk, C. Chen, N. Jawale, A. V. Kurutukulam, M. A. Lin, A. Mandlekar, K. Patzwaldt, J. Welsh, H. Zhao, F. Anes, J.-F. Lafleche, N. Moënné-Loccoz, S. Park, R. Stepinski, D. V. Gelder, C. Amevor, J. Carius, J. Chang, A. H. Chen, P. de Heras Ciechomski, G. Daviet, M. Mohajerani, J. von Muralt, V. Reutsky, M. Sauter, S. Schirm, E. L. Shi, P. Terdiman, K. Vilella, T. Widmer, G. Yeoman, T. Chen, S. Grizan, C. Li, L. Li, C. Smith, R. Wiltz, K. Alexis, Y. Chang, D. Chu, L. J. Fan, F. Farshidian, A. Handa, S. Huang, M. Hutter, Y. Narang, S. Pouya, S. Sheng, Y. Zhu, M. Macklin, A. Moravanszky, P. Reist, Y. Guo, D. Hoeller, and G. State. Isaac lab: A gpu-accelerated simulation framework for multi-modal robot learning. *arXiv preprint arXiv:2511.04831*, 2025. URL <https://arxiv.org/abs/2511.04831>.

A Additional Training Details

Table 2: Reward designs for the RL training.

Penalty			
Term	Weight	Term	Weight
Torque limits	-0.01	Termination	-1.0
DoF position limits	-5.0	DoF velocity limits	-5.0
Torque	-1×10^{-5}	Action rate	-0.05
DoF acceleration	-2.5×10^{-7}	DoF velocity	-1×10^{-3}
Power	-1×10^{-4}	Feet max contact force	-0.1
Collision	-2.0	Slippage	-0.1
Task Reward			
Term	Weight	Term	Weight
Body pos tracking	1.0	Body vel tracking	1.0
Body ang-vel tracking	1.0	Body rot tracking	1.0

Table 3: Domain randomization parameters.

Domain Rand Term	Range	Type	Domain Rand Term	Range	Type
Friction	[0.3, 1.6]	Scale	Action delay	[0, 3]	-
Body link mass	[0.5, 1.5]	Scale	Hands link mass(kg)	[0, 1.2]	Add
Torso link mass(kg)	[0, 2]	Add	Torso link Com(m)	[-0.05, 0.05]	Add
Body link CoM(m)	[-0.03, 0.03]	Add	Joint armature	[0.75, 1.25]	Scale
Joint stiffness	[0.75, 1.25]	Scale	Joint damping	[0.75, 1.25]	Scale
Push robot base(m/s)	[-0.25, 0.25]	Add	Push robot hands(N)	[-2.0, 2.0]	Add

B Motion Skeletons Details

In the construction of the motion dataset, we first use the human SMPL skeleton to standardize all collected human motions. This step converts heterogeneous motion data from different sources into a unified skeletal representation. After standardization, we perform shape fitting to map the human motions onto a humanoid SMPL skeleton, which is designed to match the morphology of the humanoid robot. The resulting humanoid motions are used as the reference targets for training the generator policy. Both SMPL models contains 23 body joints and are shown in Fig. 6

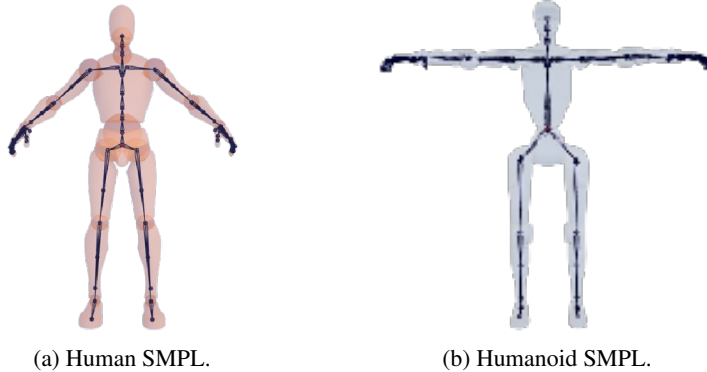


Figure 6: SMPL body models.